

ECG Signal Processing and Analysis using MATLAB

1. Introduction

Electrocardiography (ECG) is a widely used technique for monitoring the electrical activity of the heart. ECG signals provide valuable information about heart rhythm and cardiac health. However, raw ECG signals often contain noise and artifacts that can affect analysis.

This project focuses on processing and analyzing ECG signals using MATLAB. The project includes signal visualization, noise removal, peak detection, frequency-domain analysis, and statistical feature extraction.

2. Objective

The objectives of this project are:

- Visualize ECG signals
 - Remove noise from raw ECG data
 - Detect significant peaks in the ECG waveform
 - Analyze frequency components using Fast Fourier Transform (FFT)
 - Extract statistical features from the signal
 - Gain practical experience in biomedical signal processing
-

3. Tools Used

- MATLAB
 - Signal Processing Toolbox
-

4. Dataset Description

The ECG dataset consists of multiple ECG recordings.

- Columns 1–140 contain ECG signal samples.
- Column 141 contains signal labels or class information.

The ECG signals are converted into MATLAB arrays for processing and analysis.

5. Methodology

Step 1: Data Loading

The ECG dataset is imported into MATLAB and converted into an array format.

```
data = table2array(ecg);  
signals = data(:,1:140);  
labels = data(:,141);
```

Step 2: ECG Signal Visualization

The first ECG sample is plotted to observe the waveform characteristics.

```
plot(signals(1,:))
```

This provides a visual representation of the raw ECG signal.

Step 3: Noise Removal

A bandpass filter is applied to remove unwanted noise.

Filter specifications:

- Lower Cutoff Frequency = 0.5 Hz
- Upper Cutoff Frequency = 40 Hz
- Sampling Frequency = 360 Hz

```
filtered_signal = bandpass(ecg_signal,[0.5 40],Fs);
```

The filter removes:

- Baseline wander
- High-frequency noise
- Motion artifacts

while preserving important ECG components.

Step 4: Signal Comparison

The original and filtered signals are plotted for comparison.

This helps evaluate the effectiveness of the filtering process and verify that useful signal information is retained.

Step 5: Peak Detection

Significant peaks are identified using MATLAB's `findpeaks()` function.

```
[peaks,locations] = findpeaks(filtered_signal,...  
'MinPeakHeight',0.5,...  
'MinPeakDistance',15);
```

Parameters used:

- Minimum Peak Height = 0.5
- Minimum Peak Distance = 15 samples

Peak detection helps identify important ECG features such as R-peaks.

Step 6: Frequency Analysis

Fast Fourier Transform (FFT) is applied to examine the frequency content of the ECG signal.

```
Y = fft(filtered_signal);
```

FFT converts the signal from the time domain to the frequency domain.

The resulting frequency spectrum helps identify dominant frequency components and evaluate signal quality.

Step 7: Statistical Feature Extraction

The following statistical features are calculated:

- Mean
- Maximum Value

- Minimum Value
- Standard Deviation

```
mean_value = mean(ecg_signal);  
max_value = max(ecg_signal);  
min_value = min(ecg_signal);  
std_value = std(ecg_signal);
```

These parameters provide quantitative information about the ECG signal characteristics.

6. Results

The project successfully generated:

- Original ECG Signal Plot
- Filtered ECG Signal Plot
- Peak Detection Visualization
- Frequency Spectrum Plot
- Statistical Feature Analysis

The filtering process improved signal quality, and peak detection successfully identified significant ECG waveform features.

7. Applications

- Biomedical Signal Processing
 - Cardiac Health Monitoring
 - ECG Feature Extraction
 - Medical Data Analysis
 - Healthcare Research
 - Machine Learning Preprocessing
-

8. Learning Outcomes

This project provided practical experience in:

- MATLAB Programming
- Digital Signal Processing
- Biomedical Signal Analysis

- Noise Filtering Techniques
 - FFT Analysis
 - Peak Detection Algorithms
 - Statistical Feature Extraction
-

9. Future Improvements

- Heart Rate Calculation
 - QRS Complex Detection
 - Arrhythmia Classification
 - Machine Learning-Based ECG Analysis
 - Real-Time ECG Monitoring
 - Automated Cardiac Anomaly Detection
-

10. Conclusion

This project successfully demonstrated the processing and analysis of ECG signals using MATLAB. Through filtering, peak detection, frequency-domain analysis, and statistical feature extraction, meaningful information was obtained from the ECG data. The project highlights the importance of signal processing techniques in biomedical engineering and healthcare applications.